CSP 571 Project Phase 2: Crypto Tweet Sentiment and Price Movement Analysis

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1 Project Overview

We aim to investigate whether Elon Musk's cryptocurrency-related tweets, especially those mentioning Dogecoin and Bitcoin, are associated with subsequent price movements in DOGE/USD and BTC/USD. The analysis will explore whether social media sentiment can act as a short-term or medium-term indicator for cryptocurrency returns.

2 Data Integration Steps

1. Merge Sentiment and Market Data The primary merge combined the daily sentiment dataset (`sentiment_classified.csv`) with Bitcoin price data (BTC-USD from Yahoo Finance) using matching dates (`createdAt` → `date`). This alignment linked each tweet's sentiment score to the corresponding Bitcoin market performance on the same day.

2. Filter for Bitcoin-Relevant Tweets After merging, only rows where the `category` was "Bitcoin" were retained. Tweets related to other cryptocurrencies or unrelated topics were removed to focus analysis on Bitcoin-specific sentiment.

3. Clean and Simplify Dataset Unnecessary columns (non-relevant flags, usernames, unrelated keyword fields) were dropped to create a compact analysis-ready dataset containing:

- **date / createdAt**
- **sentiment (normalized 0–10)**
- **BTC returns (1D, 7D, 30D)**
- **Tweet volume / count per day**

This refined dataset, referred to as `btc_sentiment_daily`, formed the basis for all subsequent analysis and modeling.

3 Exploratory Data Analysis

To investigate how Twitter activity relates to Bitcoin market dynamics, we conducted a detailed exploratory data analysis (EDA) incorporating tweet sentiment, tweet volume, and subsequent BTC price behavior. The goal was to visually and statistically inspect whether signals from tweets align with short-horizon Bitcoin market movements.

3.1 Interactive Tweet-Level Dashboard

An interactive dashboard was created to facilitate rapid exploration of tweet-specific effects. Users can filter by Tweet ID and view the corresponding:

- Tweet content and timestamp
- Bitcoin closing price over the following 30 days
- Computed next-day, 7-day, and 30-day returns

This tool helps identify patterns across individual tweets and their short-term market impact.

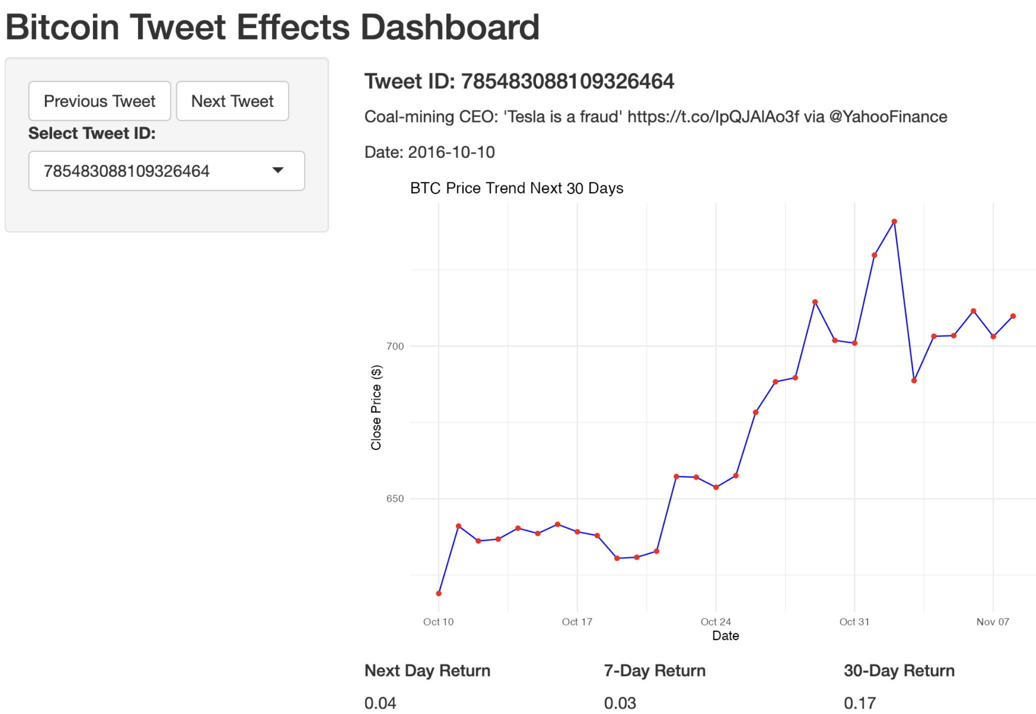

Figure 1: Interactive dashboard showing tweet details and subsequent 30-day BTC price trend.

3.2 Sentiment vs. Next-Day Bitcoin Return

Daily aggregated sentiment scores were compared against the next-day Bitcoin return to assess whether shifts in sentiment correspond with measurable price reactions.

A scatter plot with a fitted regression line (Figure 2) indicates a slight positive relationship, suggesting that more positive tweet sentiment may be associated with marginally higher next-day BTC returns.

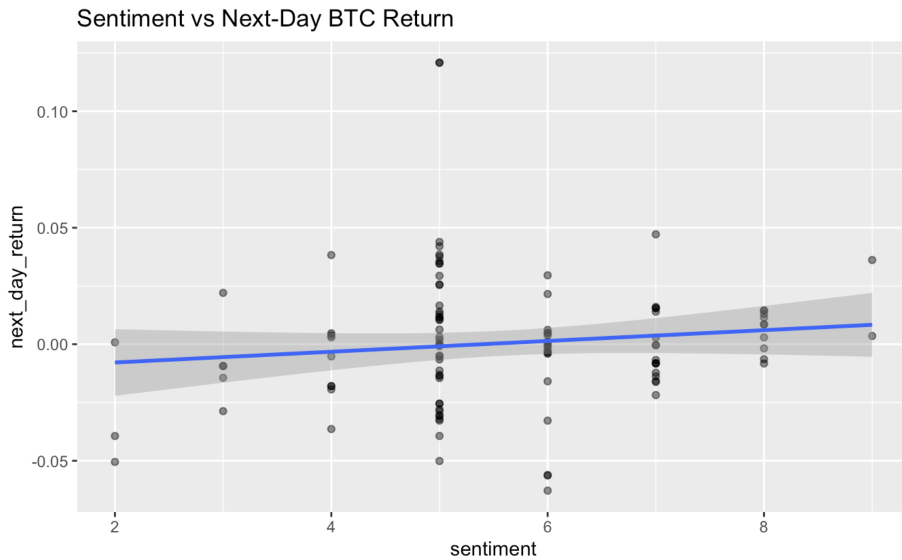

Figure 2: Relationship between tweet sentiment and next-day BTC returns.

3.3 Tweet Volume vs. BTC Volatility

We also examined whether days with higher tweet volume corresponded to greater short-term Bitcoin price volatility. The results, shown in Figure 3, exhibit a weak positive trend, indicating that increased tweet activity may be loosely associated with heightened BTC volatility.

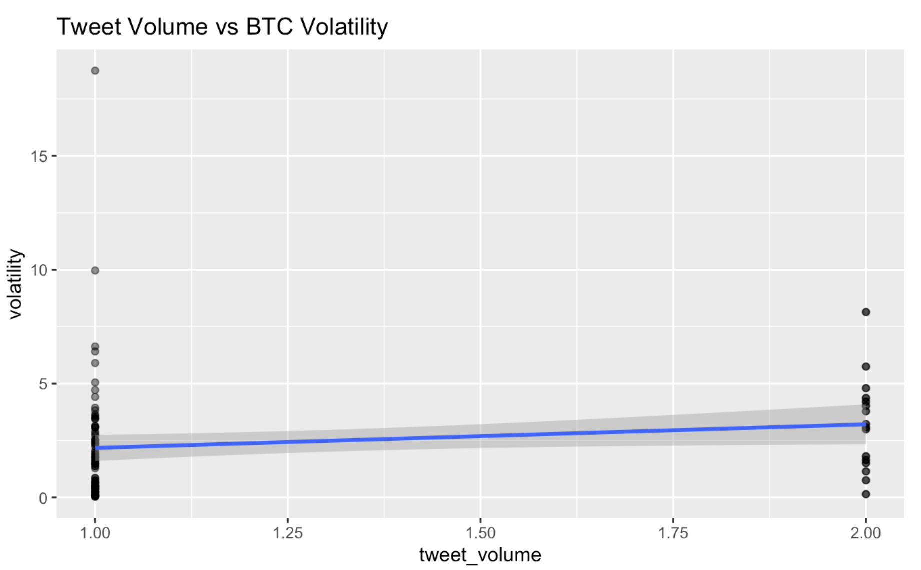

Figure 3: Tweet volume versus short-term BTC price volatility.

4 Key Research Questions

4.1 Descriptive Questions

- How does the overall sentiment about Bitcoin vary over time?
- Do peaks in tweet volume correspond to high volatility or large BTC price changes?

4.2 Analytical Questions and Hypotheses

Hypothesis	**Null Hypothesis**	**Test / Approach**
H1: Positive tweet sentiment is associated with positive next-day BTC returns.	Tweet sentiment has no relationship with next-day BTC returns.	Correlation and regression between sentiment polarity and next-day return.
H2: Negative tweets predict short-term declines in Bitcoin prices (1–7 days).	Sentiment does not influence short-term returns.	Event study analysis and t-tests of mean returns after negative tweets.
H3: Periods with high tweet activity have higher BTC volatility.	Tweet volume is unrelated to BTC volatility.	Regression of volatility on tweet volume.
H4: Sentiment effects weaken over time (i.e., 30-day return correlation $<$ 7-day).	Sentiment's effect is consistent over time.	Compare R^2 or effect sizes across time horizons.

Table 1: Summary of Analytical Hypotheses and Testing Approaches

5 Analytical and Modeling Techniques

5.1 Data Transformation

- Convert categorical sentiment to numeric values (e.g., +1, 0, -1).
- Aggregate daily tweet metrics:
  - Mean sentiment per day
  - Tweet volume per day

- Merge with daily BTC data to create a combined dataset, denoted as `btc_sentiment_daily`.

5.2 Modeling Framework

5.2.1 (a) Correlation & Regression Analysis

- Compute Pearson and Spearman correlations between:
  - Daily mean sentiment and next-day returns
  - Daily mean sentiment and 7-day returns
  - Daily mean sentiment and 30-day returns
- Fit regression models of the form:

$$\text{Return}_{t+h} = \beta_0 + \beta_1 \,\text{Sentiment}_t + \beta_2 \,\text{Volume}_t + \epsilon_t$$

  where $h = 1, 7, 30$ represents different forward-looking time horizons.

6 Deliverables

- **Statistical Analysis Report:** Hypothesis testing and regression results.
- **Predictive Model:** Forecast or classification model evaluating the predictive power of sentiment.